Certified Prediction Stability for One-Dimensional Decision Trees: A Discontinuity-Cover Approach

Mohammed Amine Chaira

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Abstract

For a one-dimensional decision tree under uniform input noise, the prediction-change probability is the integral of a binary step function. We construct an exact range oracle that retains endpoint values and removes tree boundaries inactive in the binary response. A boundary-occupancy profile gives the uniform certificate radius and an adaptive query bound of $O(b \log(b/\varepsilon))$ for $b > 0$ active boundaries. Independently uncertain, ordered thresholds produce a sharp probability interval with a positive uncertainty floor. Freezing cells inside uncertainty bands preserves this floor while avoiding ineffective refinement; one explicit example separates logarithmic from reciprocal query growth. When uncertainty can change the nominal prediction, conditioning on at most two parameter boxes gives the sharp closed probability hull, including unattained limiting values. An example shows that unconditioned pointwise ranges can remain uninformative at every mesh size. Exact rational experiments illustrate the results on finite step models and a one-feature Iris tree. Direct region sums provide exact reference probabilities throughout.

Keywords: decision trees; prediction stability; interval enclosures; adaptive quadrature; threshold uncertainty.

1 Introduction

A measured input can cross a decision threshold when perturbed. Given a noise distribution, the probability of a changed prediction describes the stability of a fixed classifier near that input. We seek rational bounds

$$p \in [p^-, p^+], \qquad \frac{p^+ - p^-}{2} \le \varepsilon, \tag{1.1}$$

so that the midpoint has absolute numerical error at most ε.

For a tree, this probability can be calculated by summing the probabilities of its prediction regions. Schweimer and Scher [1] develop such region-based calculations for tree classifiers under probabilistic distortions. Formal robustness verification for tree ensembles is also studied through abstract interpretation by Ranzato and Zanella [2]. Here the input and the perturbation law are one-dimensional, and the object of study is an interval-sum certificate and the number of range queries needed to obtain it.

The underlying covering argument is simple. Regions on which the binary response is constant contribute exactly; the total length of unresolved regions bounds the remaining uncertainty. This is the interval-cover viewpoint of certified quadrature [3]. The definitions and proofs needed for the finite-tree setting are given below. They also distinguish numerical overestimation from uncertainty about threshold locations, which cannot be removed by subdivision alone.

2 Prediction change as a step-function integral

Let $T : [0,1] \to \mathcal{Y}$ be a finite classification tree, where $\mathcal{Y}$ is a finite label set. At a threshold t, route $z \le t$ left and $z > t$ right. Assume each threshold lies strictly inside the interval reaching its node. The leaves then give an ordered partition into positive-length intervals, with left endpoints excluded and right endpoints included, except that the first leaf includes zero.

Fix rational x, δ with $\delta > 0$ and $[x-\delta, x+\delta] \subseteq [0,1]$. For U uniform on $[0,1]$, put

$$Z = x - \delta + 2\delta U, \qquad y_\star = T(x), \qquad g(u) = \mathbf{1}_{\{T(x-\delta+2\delta u) \neq y_\star\}}. \tag{2.1}$$

Then

$$p = \mathbb{P}(T(Z) \neq T(x)) = \int_0^1 g(u)\, du. \tag{2.2}$$

This quantity compares the perturbed prediction with the nominal prediction. It does not require a true class label, and its interpretation is conditional on the chosen noise law.

For example, if $T(z) = \mathbf{1}_{\{z>1/2\}}$, $x = 12/25$, and $\delta = 1/20$, then the noise interval is $[43/100, 53/100]$. The transformed threshold is $7/10$, giving $p = 3/10$.

2.1 Active boundaries and endpoint values

Transform thresholds inside the noise interval by

$$t \longmapsto \frac{t - (x-\delta)}{2\delta}.$$

Evaluate the binary response between consecutive transformed thresholds and merge adjacent regions with the same response. A transition between two labels both different from $y_\star$ disappears in this operation.

Write the retained interior boundaries as $0 < t_1 < \cdots < t_s < 1$, set $t_0 = 0$, $t_{s+1} = 1$, and let $c_j \in \{0,1\}$ be the response on $(t_j, t_{j+1}]$, $0 \le j \le s$. Consecutive c_j differ. Store $e_0 = g(0)$ and $e_1 = g(1)$ separately, overriding the interval description at those endpoints. The active boundary set is

$$\mathcal{A} = \{t_1, \ldots, t_s\} \cup \mathcal{A}_0 \cup \mathcal{A}_1, \qquad b = \#\mathcal{A}, \tag{2.3}$$

where $\mathcal{A}_0 = \{0\}$ when $e_0 \neq c_0$ and is empty otherwise, with the analogous definition at one. The tree convention gives $e_1 = c_s$, but e_0 can differ from c_0 when the noise interval begins at a tree threshold. These isolated values do not affect the integral; they do affect pointwise ranges on closed cells.

Proposition 2.1 (Integrability and direct summation)**.** *The response g is Riemann integrable, and*

$$p = \sum_{j=0}^{s} c_j (t_{j+1} - t_j). \tag{2.4}$$

Equivalently, if the original leaf intervals are I_k with labels y_k, then

$$p = \frac{1}{2\delta} \sum_{k : y_k \neq T(x)} |I_k \cap [x-\delta, x+\delta]|. \tag{2.5}$$

Both expressions are rational when the thresholds, x, and δ are rational.

Proof. The discontinuity set (2.3) is finite and can be covered by intervals of arbitrarily small total length. Outside these intervals the response is locally constant, so the Darboux gap tends to zero under refinement. Alternatively, split at the finitely many boundaries and integrate the constant pieces. This gives (2.4). Summing the lengths of the original leaf intersections gives (2.5); endpoint conventions do not change their lengths. □

Formula (2.5) solves the explicit one-dimensional problem exactly and serves as the reference computation in the experiments. The adaptive analysis concerns a different representation of the result: an enclosure assembled from local ranges.

The same construction applies to disagreement,

$$g(u) = \mathbf{1}_{\{T_1(u) \neq T_2(u)\}}. \tag{2.6}$$

Merge the two threshold lists, compare labels on each region, and remove inactive boundaries. Intersections of the original leaf regions give an independent direct sum.

3 Range certificates

Assume rational tree thresholds. For $v \in [0,1]$, define $k(v) = \#\{i : t_i < v\}$. On a positive-length closed cell $J = [a, d]$, form

$$V(J) = \{c_{k(a)}, \ldots, c_{k(d)}\} \cup \{e_j : j \in J \cap \{0,1\}\}. \tag{3.1}$$

The oracle returns $R(J) = [\min V(J), \max V(J)]$. Rational binary searches determine the two indices, so this is a finite exact procedure.

Proposition 3.1 (Exact pointwise ranges)**.** *For every positive-length closed $J \subseteq [0,1]$, the oracle is exactly $[\min_{u \in J} g(u), \max_{u \in J} g(u)]$ and is nested under interval inclusion.*

Proof. At an interior threshold, $k(t_i) = i - 1$, which gives the left branch value. Every branch indexed from $k(a)$ to $k(d)$ occurs in J, possibly only at a for the first branch. If a domain endpoint lies in J, its adjacent branch still occurs immediately inside the interval, and its actual endpoint value is added separately. Thus $V(J)$ is the set of attained values. Nesting follows from inclusion of these sets. □

For a partition P into closed intervals with disjoint interiors and union $[0,1]$, write $R(J) = [l_J, u_J]$ and set

$$A(P) = \sum_{J \in P} |J| l_J, \quad B(P) = \sum_{J \in P} |J| u_J, \quad E(P) = \frac{B(P) - A(P)}{2}. \tag{3.2}$$

A cell with range $[0,1]$ is *ambiguous*; all others are resolved.

Algorithm 3.2 (Largest-radius refinement)**.** Given rational $\varepsilon > 0$, query the root $[0,1]$. While $E(P) > \varepsilon$, bisect a leaf maximizing $|J|(u_J - l_J)/2$, query both children, and update (3.2). Return $[A(P), B(P)]$ when the radius test holds.

Ties are broken by smaller depth, then smaller left endpoint. A run interrupted before the stopping test passes retains a valid enclosure without a guarantee at the requested tolerance.

Proposition 3.3 (Validity and query count)**.** *Every finite partition constructed by Algorithm 3.2 encloses p, and its midpoint error is at most $E(P)$. Refinement gives nested enclosures. After q bisections there are $q + 1$ leaves and $2q + 1$ range calls.*

Proof. Integrate the range inequalities on each cell and add. A child's range is contained in its parent's range and its length is half the parent's length. The lower sum therefore increases and the upper sum decreases. Each split adds one leaf and two calls to the initial root call. □

For a binary response, the certificate width is exactly the total length of ambiguous leaves. This identifies the unresolved part of the computation with a cover of the active boundaries.

4 Boundary occupancy and query complexity

Let $\mathcal{D}_m$ be the complete dyadic grid with $N = 2^m$ cells. Assign each active boundary to the cell with index

$$\mathrm{own}_m(t) = \min\{\lfloor 2^m t \rfloor, 2^m - 1\}, \qquad \nu_m = \#\{\mathrm{own}_m(t) : t \in \mathcal{A}\}. \tag{4.1}$$

A grid-aligned interior jump belongs to the cell on its right: that cell contains the left value at the jump and the right value immediately after it.

Lemma 4.1 (Exact ambiguity count)**.** *The ambiguous cells of $\mathcal{D}_m$ are exactly those owning a point of $\mathcal{A}$. Thus*

$$\nu_m \le \min\{2^m, b\}, \qquad E(\mathcal{D}_m) = \frac{\nu_m}{2^{m+1}}. \tag{4.2}$$

If $b > 0$, then $\nu_m = b$ for all sufficiently large m.

Proof. A jump strictly inside a cell gives both response values there. At a grid edge, the right cell contains both values and the left cell gains no variation from that jump alone. An endpoint discrepancy gives both values in its adjacent cell. Conversely, a cell owning no boundary has constant response, including its endpoints. Each ambiguous cell contributes 2^{-m-1} to the radius. Finally, sufficiently short cells cannot own two distinct points of the finite set $\mathcal{A}$. □

Theorem 4.2 (Finite-boundary query bound)**.** *If $b = 0$, Algorithm 3.2 stops after one call. If $b > 0$, $0 < \varepsilon < 1/2$, and $L \ge 1$ satisfies $\nu_L/2^{L+1} \le \varepsilon$, then*

$$Q(\varepsilon) \le 1 + 2\sum_{m=0}^{L-1} \nu_m, \qquad N_{\mathrm{ad}}(\varepsilon) \le 1 + \sum_{m=0}^{L-1} \nu_m. \tag{4.3}$$

One may take

$$L = \left\lceil \log_2 \frac{b}{2\varepsilon} \right\rceil, \qquad Q(\varepsilon) \le 1 + 2bL. \tag{4.4}$$

For a fixed nonconstant response, the smallest acceptable full uniform dyadic grid has $\Theta(b/\varepsilon)$ cells as $\varepsilon \downarrow 0$.

Proof. If $b = 0$, the response is constant, including at the endpoints. Otherwise every nonzero local radius at depth m equals 2^{-m-1}, so the priority rule processes ambiguous cells at coarser depths first. Every ambiguous node has ambiguous ancestors. Splitting all ambiguous nodes above depth L therefore gives exactly $\sum_{m<L} \nu_m$ internal nodes, with the same integral sums as $\mathcal{D}_L$: constant subtrees need no refinement. Their radius is at most ε. Greedy refinement completes this comparison tree before processing any deeper node unless it stops earlier. Proposition 3.3 gives (4.3), and $\nu_L \le b$ gives (4.4).

For sufficiently fine uniform grids, Lemma 4.1 gives $E(\mathcal{D}_m) = b/(2N)$. All coarser grids have positive radius, so the first acceptable grid is eventually among the fine grids. Rounding $b/(2\varepsilon)$ up to a power of two changes it by a factor below two. □

Clustering reduces the occupancy sum even when the total number of boundaries is unchanged. The algorithm needs no estimate of their minimum separation.

The uniform order in Theorem 4.2 is specific to pointwise ranges on closed dyadic cells. Essential ranges and direct threshold sums use different information. Query counts exclude training, extraction of the response, and arithmetic inside each call. A heap uses $O(N \log N)$ comparisons for N leaves, while rational arithmetic costs depend on the integer lengths.

5 Uncertain thresholds

Suppose the binary branch values $c_0, \dots, c_s$ are fixed, with nonzero jumps $a_i = c_i - c_{i-1}$, and the thresholds satisfy

$$0 < \ell_1 \le r_1 < \ell_2 \le r_2 < \cdots < \ell_s \le r_s < 1, \qquad \theta_i \in [\ell_i, r_i]. \tag{5.1}$$

Every coordinate may vary independently. The bands remain ordered for every admissible choice. Define g_θ using these thresholds, the left-branch convention, and adjacent branch values at zero and one. Put

$$\mathcal{U} = \bigcup_{i=1}^{s} [\ell_i, r_i], \qquad \Lambda = \sum_{i=1}^{s} (r_i - \ell_i), \qquad \rho = \Lambda/2. \tag{5.2}$$

Degenerate bands are allowed. With $s = 0$, the response is constant.

For a classifier interpretation, the ordered leaf labels and reference class remain fixed while the thresholds move. If the nominal coordinate lies outside $\mathcal{U}$, its predicted class is fixed as well. Section 5.3 treats the case in which that class can change.

Theorem 5.1 (Sharp probability range)**.** *Under* (5.1)*,*

$$p(\theta) = \int_0^1 g_\theta(u)\, du = c_s - \sum_{i=1}^{s} a_i \theta_i. \tag{5.3}$$

Let $\theta_i^- = r_i$ if $a_i > 0$ and $\theta_i^- = \ell_i$ if $a_i < 0$, and set $\theta_i^+ = \ell_i + r_i - \theta_i^-$. The attained extrema are

$$p_\star^- = c_s - \sum_i a_i \theta_i^-, \qquad p_\star^+ = c_s - \sum_i a_i \theta_i^+, \qquad p_\star^+ - p_\star^- = \Lambda. \tag{5.4}$$

Every interval containing all admissible probabilities has radius at least ρ.

Proof. Integrate $g_\theta(u) = c_0 + \sum_i a_i \mathbf{1}_{\{u > \theta_i\}}$ to obtain (5.3). Each coordinate enters affinely, so its sign determines the minimizing and maximizing endpoint. Independence and strict ordering of the bands make both corner choices feasible. Since $|a_i| = 1$, the width is $\sum_i (r_i - \ell_i) = \Lambda$. □

These extrema describe an explicitly supplied model family. Statistical calibration of the bands, overlapping bands, or correlated threshold choices require additional assumptions. With the complete independent box available, (5.4) computes its probability range directly in s terms.

5.1 Refinement above the uncertainty floor

Use the conservative family oracle

$$R_{\mathcal{U}}(J) = \begin{cases} [0,1], & J \cap \mathcal{U} \neq \varnothing, \\ [c,c], & J \cap \mathcal{U} = \varnothing, \end{cases} \tag{5.5}$$

where c is the common branch value on J. This nested oracle contains every point value of every admissible model. Subdividing a cell wholly inside a positive-length band leaves its total interval-sum contribution unchanged.

Algorithm 5.2 (Freeze band interiors)**.** Maintain the sums over all leaves. Freeze a cell if its range is a singleton or it lies wholly inside one band, retaining its contribution in both sums. Among eligible leaves, bisect one of largest local radius. Stop when the radius of the full interval sum is at most ε.

Theorem 5.3 (Certification above the floor)**.** *For rational data in* (5.1) *and $\varepsilon = \rho + \eta$ with rational $\eta > 0$, Algorithm 5.2 terminates and returns a nested enclosure containing every $p(\theta)$. If $s > 0$, set*

$$L = \max\left\{0, \left\lceil \log_2 \frac{s}{\eta} \right\rceil\right\}. \tag{5.6}$$

At most $1 + 8sL$ range calls are needed. For $s = 0$, one call suffices.

Proof. Integrating each family range proves validity for every θ. Nesting and the $2q + 1$ call count hold as before; freezing never removes uncertainty from the sums.

An eligible cell meets a band without being contained in one, so it contains a band endpoint. At each depth the $2s$ endpoints belong to at most $4s$ closed cells. All eligible cells have range width one, and the priority rule processes their coarser depths first. Splitting eligible nodes above depth L therefore requires at most $4sL$ bisections.

Frozen subtrees have the same contributions as all their descendants. The comparison tree thus has the same sums as the full depth-L grid for (5.5). A band of length $r_i - \ell_i$ is covered there by cells of total length at most $r_i - \ell_i + 2 \cdot 2^{-L}$. Consequently,

$$E(\mathcal{D}_L) \le \frac{\Lambda}{2} + s2^{-L} \le \rho + \eta. \tag{5.7}$$

The greedy run completes this tree before splitting a finer eligible cell unless it stops earlier. This gives $1 + 8sL$ calls. If $L = 0$, the root already satisfies the bound. □

The guarantee is strictly above the floor. Below it, Theorem 5.1 rules out the requested common enclosure. Equality may or may not be reached by a finite dyadic partition.

5.2 A separation between refinement rules

Proposition 5.4. *Let $g_\theta(u) = \mathbf{1}_{\{u \le \theta\}}$ with $\theta \in [1/6, 1/3]$. Its probability range is $[1/6, 1/3]$ and $\rho = 1/12$. Using* (5.5)*, ordinary largest-radius refinement needs $\Theta(\eta^{-1})$ calls to reach radius $\rho + \eta$, whereas Algorithm 5.2 needs $\Theta(\log(1/\eta))$ calls as $\eta \downarrow 0$.*

Proof. For $N = 2^m$, $m \ge 1$, the ambiguous cells cover from $\lfloor N/6 \rfloor / N$ to $(\lfloor N/3 \rfloor + 1)/N$. Since N is congruent to two or four modulo six,

$$E(\mathcal{D}_m) - \rho \in \left\{\frac{1}{3N}, \frac{2}{3N}\right\}. \tag{5.8}$$

Any dyadic partition of greatest depth d is refined by $\mathcal{D}_d$, so nesting gives $E(P) \ge E(\mathcal{D}_d)$. Reaching excess at most η requires $2^d \ge 1/(3\eta)$. Creating such a depth takes at least d splits, giving a logarithmic lower bound for both rules.

Ordinary refinement processes all ambiguous nodes at a coarser depth first. Before creating depth d, it must split all ambiguous nodes at depth $d - 2$ when $d \ge 2$. These cover length $1/6$ and number at least $2^{d-2}/6$. Hence, for sufficiently small η,

$$Q_{\text{ordinary}} \ge 2\frac{2^{d-2}}{6} \ge \frac{1}{36\eta}.$$

Conversely, complete depth $L = \lceil \log_2(2/(3\eta)) \rceil$. Its radius suffices by (5.8). At depth m, at most $2^m/6 + 2$ cells meet the band, giving at most

$$1 + 2\sum_{m<L} (2^m/6 + 2) = 1 + \tfrac{1}{3}(2^L - 1) + 4L = O(\eta^{-1})$$

ordinary calls. Theorem 5.3 supplies the matching logarithmic upper bound for frozen refinement. □

This example is a prediction-change response for a tree with labels zero to the left and one to the right of θ, using nominal coordinate $1/2$. Its nominal label is always one. The proposition compares the two subdivision rules; the affine formula still gives the exact probability range directly.

5.3 A changing nominal prediction

Let $d_0, \dots, d_s \in \mathcal{Y}$ be fixed, possibly multiclass, labels, and let h_θ be their step classifier with bands satisfying (5.1). For a fixed nominal coordinate $q \in (0,1)$, define

$$p_{\text{self}}(\theta) = \int_0^1 \mathbf{1}_{\{h_\theta(u) \neq h_\theta(q)\}}\, du. \tag{5.9}$$

The centered noise model has $q = 1/2$. A general q also covers asymmetric uniform input intervals.

For a fixed reference class y, put $c_i^y = \mathbf{1}_{\{d_i \neq y\}}$ and $a_i^y = c_i^y - c_{i-1}^y$. On a closed subbox $\mathcal{B} = \prod_i [L_i, R_i]$, define

$$F_y(\theta) = c_s^y - \sum_i a_i^y \theta_i, \tag{5.10}$$

$$\underline{F}_y(\mathcal{B}) = c_s^y - \sum_{a_i^y > 0} a_i^y R_i - \sum_{a_i^y < 0} a_i^y L_i, \tag{5.11}$$

$$\overline{F}_y(\mathcal{B}) = c_s^y - \sum_{a_i^y > 0} a_i^y L_i - \sum_{a_i^y < 0} a_i^y R_i. \tag{5.12}$$

Zero jumps contribute nothing and their bands can be removed for fixed-reference refinement.

Theorem 5.5 (Sharp hull with a changing nominal class)**.** *Construct a collection $\mathcal{H}$ of closed boxes paired with reference classes as follows.*

(i) If $q \notin \mathcal{U}$, let $k = \#\{i : r_i < q\}$ and take $\mathcal{H} = \{(\prod_i [\ell_i, r_i], d_k)\}$.

(ii) If $q \in [\ell_j, r_j]$, this band is unique. Include $(\mathcal{B}_{\geq}, d_{j-1})$, where its jth interval is replaced by $[q, r_j]$. If $\ell_j < q$, also include $(\mathcal{B}_{<}, d_j)$, where that interval is replaced by $[\ell_j, q]$. The latter box is the closure of the strict side $\theta_j < q$.

The smallest closed interval containing all admissible self-referenced probabilities is

$$[p_{\text{self}}^-, p_{\text{self}}^+] = \left[\min_{(\mathcal{B},y)\in\mathcal{H}} \underline{F}_y(\mathcal{B}),\ \max_{(\mathcal{B},y)\in\mathcal{H}} \overline{F}_y(\mathcal{B}) \right]. \tag{5.13}$$

Proof. Integration with a fixed reference class gives (5.10). Independent affine minimization and maximization give (5.11)–(5.12), including zero coefficients.

If $q \notin \mathcal{U}$, exactly k thresholds lie strictly below q, so its class is d_k for every model. Otherwise all thresholds before j lie below q and all those after j lie above it. For $\theta_j \geq q$, the left-branch convention gives $h_\theta(q) = d_{j-1}$; for $\theta_j < q$, it gives $h_\theta(q) = d_j$.

On each side, p_{self} is the corresponding affine function. The first side is closed. The second exists exactly when $\ell_j < q$, and is dense in $\mathcal{B}_{<}$. Continuity of its fixed-class affine function therefore makes its infimum and supremum equal the extrema on that closure. Taking the smaller infimum and larger supremum gives (5.13). Every closed enclosure must contain those two limits. □

The endpoints are an infimum and supremum; a strict-side endpoint need not be attained. At most two affine box calculations, each with s terms, give the exact hull. Its radius is $\rho_{\text{self}} = (p_{\text{self}}^+ - p_{\text{self}}^-)/2$.

Corollary 5.6 (Conditional refinement)**.** *Assume rational bands and q, and fix rational $\eta > 0$. Index $\mathcal{H}$ by h, let s_h be the number of nonzero jumps for reference class y_h, and put*

$$\rho_h = \frac{\overline{F}_{y_h}(\mathcal{B}_h) - \underline{F}_{y_h}(\mathcal{B}_h)}{2}.$$

Remove zero-jump bands from each component and apply Algorithm 5.2 at tolerance $\rho_h + \eta/2$, obtaining $[A_h, B_h]$. Then

$$\left[\min_h A_h, \max_h B_h\right] \tag{5.14}$$

contains every $p_{\mathrm{self}}(\theta)$ and has radius at most $\rho_{\mathrm{self}} + \eta$. With

$$L_h = \max\{0, \lceil \log_2(2s_h/\eta) \rceil\} \quad (s_h > 0), \qquad L_h = 0 \quad (s_h = 0),$$

the total number of calls is at most

$$\sum_h (1 + 8s_h L_h) \le 2 + 16s \max\{0, \lceil \log_2(2s/\eta) \rceil\} \qquad (s > 0). \tag{5.15}$$

Decreasing the excess budget with the fixed tie rule gives nested output intervals. If $s = 0$, the response is zero and one root call suffices.

Proof. Each active binary jump has magnitude one. Its component sharp width is $\sum_i |a_i^{y_h}|(R_i - L_i) = 2\rho_h$, so Theorem 5.3 applies. Let l_h, u_h be its sharp endpoints. Validity and the stopping test give

$$0 \le (l_h - A_h) + (B_h - u_h) = (B_h - A_h) - (u_h - l_h) \le \eta.$$

Thus $A_h \ge l_h - \eta$ and $B_h \le u_h + \eta$. Taking the hull proves inclusion and the radius bound in (5.14). The closure of a strict side adds only limiting fixed-class probabilities, which any closed common enclosure must contain.

The component excess is $\eta/2$, giving the first query bound. There are at most two components with $s_h \le s$, giving the second. A run at a smaller tolerance extends a prefix of the same deterministic refinement sequence. Each component interval is therefore nested, and taking their hull preserves nesting. □

Proposition 5.7 (Unconditioned ranges can lose parameter dependence)**.** *Let $h_\theta(u) = \mathbf{1}_{\{u>\theta\}}$, $\theta \in [2/5, 3/5]$, and $q = 1/2$. The sharp range of p_{self} is $[2/5, 1/2]$. Nevertheless, on every positive-length closed interval J, the exact range of $\mathbf{1}_{\{h_\theta(u) \ne h_\theta(q)\}}$ over all $u \in J$ and admissible θ is $[0, 1]$. Every unconditioned interval-sum certificate is therefore $[0, 1]$.*

Proof. For $\theta < 1/2$, the nominal class is one and the probability is θ. For $\theta \ge 1/2$, the nominal class is zero and the probability is $1 - \theta$. Their union is $[2/5, 1/2]$.

At $u = q$, every response is zero. If $u < q$, choose $\theta = 3/5$ for response zero and $\theta = \max\{2/5, u\} < q$ for response one. If $u > q$, choose $\theta = 2/5$ for response zero and $\theta = q$ for response one. Every positive-length J contains a point different from q, so its family range is $[0, 1]$. The interval sums over a domain of length one remain $[0, 1]$. □

The two predictions in (5.9) share the same threshold parameter. Unconditioned pointwise ranges discard that dependence. Conditioning retains enough of it to recover the sharp hull.

6 Exact rational experiments

All thresholds, range bounds, interval sums, priorities, and stopping tests use Python's `Fraction`. Displayed decimal endpoints are rounded outward after the exact certificate has been obtained. The accompanying script regenerates the data and figures from the supplied inputs.

6.1 Fixed models

The ordinary suite has nine responses. A single threshold and a narrow positive band use boundaries $7/10$ and $3699/10000, 3701/10000$, respectively. Two alternating binary models use the twelve separated thresholds $k/13$, $1 \le k \le 12$, and twelve clustered thresholds $37/100 + k/100000$, $0 \le k < 12$. A multiclass model has thresholds $1/5, 2/5, 3/5, 4/5$ and labels $(0, 1, 2, 1, 0)$, with reference class two. A disagreement response compares trees with labels $(0, 1, 0)$ and thresholds $(1/3, 2/3)$ and $(2/5, 3/4)$. Its exact probability is $3/20$. The remaining cases are an Iris response, a zero response with an isolated value one at the left endpoint, and a constant response.

The Iris example uses the 150 petal-width observations and labels distributed with scikit-learn 1.8.0 [4, 5]. The finite decimals are treated as exact inputs, normalized by

$$z = \frac{\text{petal width in cm} - 1/10}{12/5}. \tag{6.1}$$

The trainer uses rational Gini gains, a depth cap of four, and minimum leaf size four. Candidate thresholds are midpoints of distinct feature values; gain ties select the leftmost candidate, and majority ties select the smallest class label. The resulting threshold list and leaf labels are

$$(7/24, 25/48, 29/48, 11/16, 35/48), \qquad (0, 1, 1, 1, 2, 2).$$

At $x = 2/3$ and $\delta = 1/8$, the exact prediction-change probability is $5/12$. Three raw thresholds lie in the noise interval, but only one changes the binary response. In measurement units, the nominal width is 1.7 cm and the prescribed perturbation is uniform within ± 0.3 cm. This experiment certifies the fitted model under that noise law; it does not estimate classification accuracy or measurement noise from the data.

Table 1: Selected ordinary certificates at $\varepsilon = 2^{-14}$. Bounds are rounded outward to seven decimal places. N_{unif} is the smallest acceptable full dyadic grid.

Case	$[p^-, p^+]$	N_{ad}	Q	N_{unif}
Single threshold	$[0.2999267, 0.3000489]$	14	27	8,192
Narrow band	$[0.0001220, 0.0002442]$	16	31	16,384
12 separated	$[0.4614562, 0.4615784]$	168	335	131,072
12 clustered	$[0.0000000, 0.0001221]$	14	27	8,192
Iris boundary	$[0.4166259, 0.4167481]$	14	27	8,192
Disagreement	$[0.1499633, 0.1500855]$	55	109	32,768

Each response is run at the eight tolerances $2^{-4}, 2^{-6}, \ldots, 2^{-18}$, giving 72 certificates. All contain their exact references. Uniform counts follow from the exact occupancy formula (4.2), so there is no need to enumerate the full grids. At $\varepsilon = 2^{-14}$, the separated and clustered models use 168 and 14 adaptive leaves despite having the same number of active boundaries. Figure 1 shows the occupancy profiles responsible for this difference. The endpoint example has integral zero but a nonzero closed-range radius, illustrating the role of isolated point values.

6.2 Threshold uncertainty

For the four-band model, place a band of half-width $1/1000$ around each $i/5$, $1 \le i \le 4$, with binary branch values $(0, 1, 0, 1, 0)$. The nominal coordinate $1/2$ lies outside every band, and Theorem 5.1 gives

$$[p_\star^-, p_\star^+] = [99/250, 101/250] = [0.396, 0.404], \qquad \rho = 1/250. \tag{6.2}$$

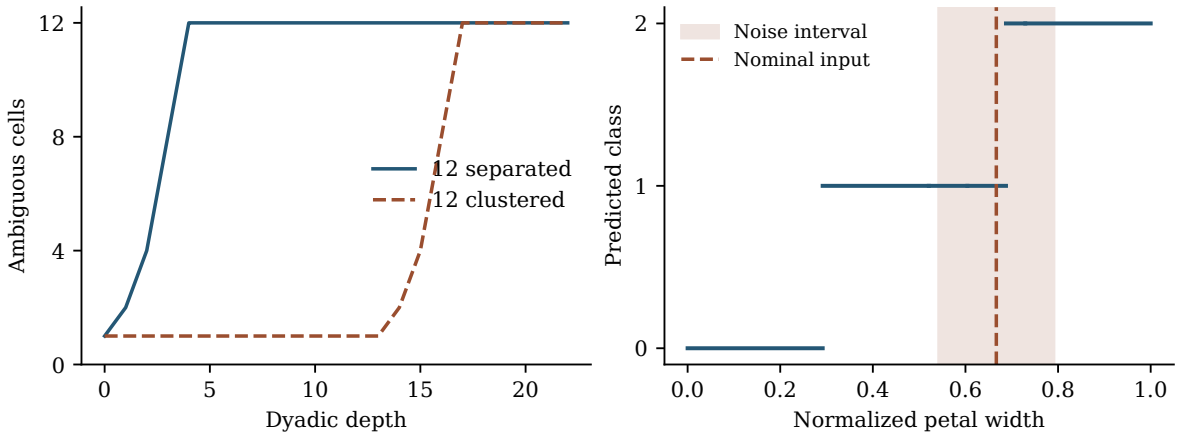


Figure 1: Left: occupancy profiles of the two twelve-boundary responses. Right: the rational Iris tree and the noise interval at $x = 2/3$, $\delta = 1/8$. Only the transition at $11/16$ changes the binary response in that interval.

Table 2 compares the two refinement rules. At excess $\eta = 2^{-16}$, ordinary refinement uses 2,211 calls and frozen refinement uses 195. For the single band in Proposition 5.4, the counts at $\eta = 2^{-15}$ are 5,491 and 55. The latter example has the asymptotic orders proved above, rather than an order inferred from the plotted points.

Table 2: Four uncertain thresholds at tolerance $1/250 + \eta$. Every interval encloses (6.2); displayed bounds are rounded outward.

η	Q_{ordinary}	Q_{frozen}	Frozen enclosure
2^{-6}	45	45	$[0.3906250, 0.4296875]$
2^{-8}	67	67	$[0.3945312, 0.4101563]$
2^{-10}	107	99	$[0.3955078, 0.4052735]$
2^{-12}	185	125	$[0.3956298, 0.4040528]$
2^{-14}	557	161	$[0.3959350, 0.4040528]$
2^{-16}	2,211	195	$[0.3959808, 0.4040070]$

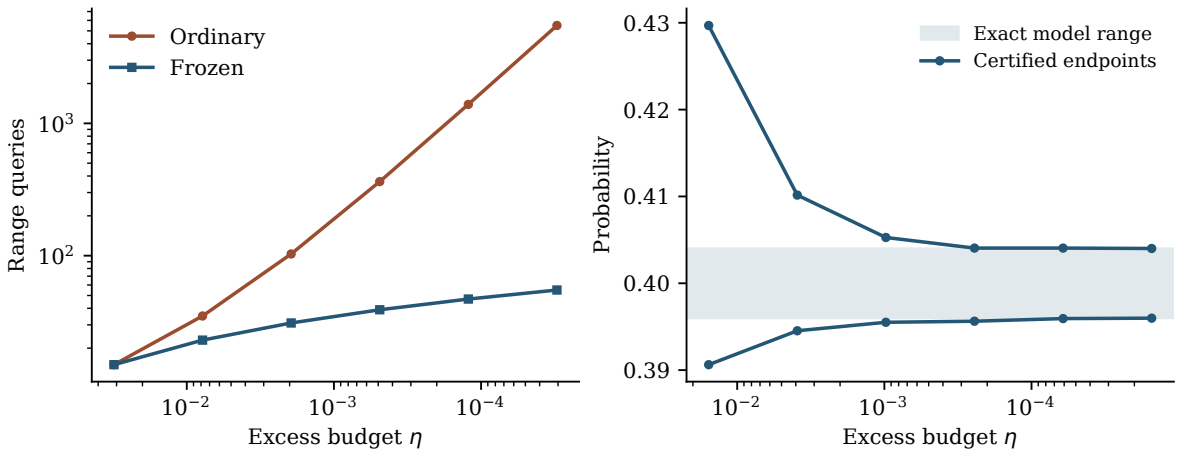


Figure 2: Left: query counts for the single band $[1/6, 1/3]$. Right: frozen enclosures for the four-band example approach its exact probability range. Interior band uncertainty remains in the sums.

6.3 Changing nominal labels

Three prescribed families test Theorem 5.5, all with $q = 1/2$.

(i) The binary model with labels $(0, 1)$ and band $[2/5, 3/5]$ is Proposition 5.7. Its exact probability range is $[2/5, 1/2]$.

(ii) For labels $(0, 1, 2)$ and bands $[1/5, 1/4]$, $[9/20, 1/2]$, the probability is θ_2 when $\theta_2 < 1/2$ and $\theta_1 + 1/2$ when $\theta_2 = 1/2$. The actual set is

$$[9/20, 1/2) \cup [7/10, 3/4], \tag{6.3}$$

whose sharp closed hull is $[9/20, 3/4]$.

(iii) For labels $(0, 1, 2, 1)$ and bands $[1/10, 3/20]$, $[2/5, 3/5]$, $[3/4, 4/5]$, the probability is $\theta_1 + \theta_3 - \theta_2$ when $\theta_2 \geq 1/2$, with range $[1/4, 9/20]$. When $\theta_2 < 1/2$, it is $1 + \theta_2 - \theta_3$, with range $[3/5, 3/4)$. The joint closed hull is $[1/4, 3/4]$, and its upper endpoint is not attained.

These examples distinguish actual probabilities from limits on a strict parameter side. Figure 3 displays the first two cases.

Table 3: Conditional certificates at $\eta = 2^{-14}$. The allowed radius is the exact hull radius plus η. Bounds are rounded outward to six decimal places. Query counts include both component roots.

Case	Exact hull	Certified hull	Q
Binary crossing	$[0.4, 0.5]$	$[0.399993, 0.500031]$	118
Multiclass endpoint	$[0.45, 0.75]$	$[0.449981, 0.750031]$	176
Multiclass crossing	$[0.25, 0.75]$	$[0.249969, 0.750031]$	286

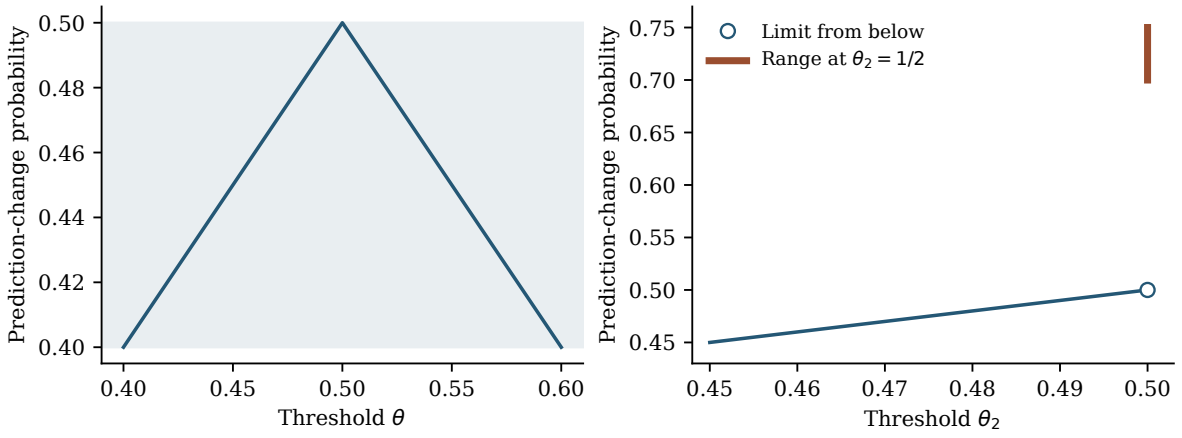


Figure 3: Left: the binary self-referenced probability in Proposition 5.7. Right: the multiclass endpoint case. Below $\theta_2 = 1/2$, the probability equals θ_2; the open circle marks its limit. At equality the nominal class changes and the probability ranges over $[0.7, 0.75]$.

Each family is run at $\eta = 2^{-6}, 2^{-8}, \ldots, 2^{-16}$, giving 18 joint enclosures and 36 component certificates. References are calculated by summing raw leaf lengths at parameter corners, including one-sided nominal limits. The script checks the hulls using an additional enumeration of threshold endpoints and nominal-side limits. All intervals contain their sharp reference hulls and meet the excess-radius tests.

6.4 Reproduction and checks

Run `python reproduce.py` to regenerate the ordinary, uncertain-threshold, and conditional certificate records, together with the three figures. The exact Iris feature data and fitted model are included. Range checks compare the oracle with direct values on every constant piece of each tested cell; occupancy checks compare boundary ownership with complete dyadic grids. Additional cases cover degenerate bands and nominal coordinates at both band endpoints. Numerical records and validation totals are written to the data directory.

7 Conclusion

For finite one-dimensional trees, active binary-response boundaries determine the interval-sum certificate. The occupancy profile gives both the exact uniform radius and an adaptive query bound, and explains why clustering can reduce refinement even when boundary counts agree.

Threshold uncertainty changes the stopping problem. Its affine probability range gives a sharp uncertainty floor; freezing band interiors avoids subdivision that cannot reduce that uncertainty. If the nominal prediction can change, conditioning on its class preserves the shared parameter dependence and reduces the calculation to at most two boxes. Direct leaf and affine sums remain exact baselines for these explicit models. The adaptive results describe how local range certificates approach the same probabilities under the stated one-dimensional noise and threshold assumptions.

A Implementation details

For a node with class counts n_y and total n, the weighted Gini criterion is

$$n\,\mathrm{Gini} = n - \frac{\sum_y n_y^2}{n}.$$

The trainer chooses the first strictly maximal positive reduction over candidate midpoints, subject to the depth and minimum-leaf constraints. The fixed tree produced by this procedure is the object certified. Its exact rational comparisons specify the prediction semantics independently of a floating-point library.

The response oracle uses two binary searches in its retained threshold list. If the indices differ, the alternating binary branches contain both zero and one; otherwise the branch is constant, subject to endpoint corrections. Direct reference integration clips the original tree's leaf intervals against the noise interval. For disagreement it intersects the two original leaf partitions.

For threshold bands, a full grid of size N has ambiguous indices

$$\max\{0, \lceil N\ell \rceil - 1\} \le j \le \min\{N-1, \lfloor Nr \rfloor\}.$$

This follows from the closed-cell intersection inequalities $(j+1)/N \ge \ell$ and $j/N \le r$. The freezing rule retains the contributions of every cell contained in a band.

Conditional refinement retains the original multiclass labels until the parameter sides are constructed. The side $\theta_j \ge q$ uses the left label, and the strict side $\theta_j < q$ uses the right label. Its closure is used only for the continuous fixed-reference calculation; it is not evaluated with the boundary model's own nominal class. Zero binary jumps are then removed separately in each component. With h components and N total component leaves, the total number of range calls is $2N - h$.

References

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[3] M. A. Chaira, *From discontinuity covers to certified adaptive quadrature*, manuscript, 2026.

[4] Scikit-learn developers, `load_iris`, API documentation. `https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_iris.html`. The experiments use the data bundled with version 1.8.0.

[5] R. A. Fisher, *Iris* [dataset], UCI Machine Learning Repository, 1988. doi:10.24432/C56C76. Distributed under CC BY 4.0. The supplied feature extract follows the corrected scikit-learn variant.